

Student Support Agent

Project Report

1. Project Overview

The Student Support Agent is a simple university support system that helps students answer common university-related questions. The system uses student information stored in a CSV file and provides answers about courses, teachers, assignment deadlines, and classrooms. It also demonstrates human approval for important actions and provides a FastAPI endpoint with basic monitoring.

2. Problem / Use Case

Students often need quick answers to common university questions such as their course, teacher, assignment deadline, or classroom. The purpose of this project is to provide these answers automatically using a simple student support agent. The system uses university information stored in a CSV file as its data source.

3. System Design

The system uses LangGraph to control the workflow. The main flow receives a student's question, searches the university information file, and generates an answer. For important actions, a human approval step is included before the action can continue.

Main flow:

Student → Student Question → LangGraph → University CSV → Answer

For important actions:

Student → Request → Human Approval → Continue / Stop

4. Framework Choice

LangGraph was selected because it is suitable for controlled multi-step workflows and state management. It allows the different steps of the Student Support Agent to be connected clearly. It also allows a human approval step to be included for important actions. CrewAI was not necessary because this project does not require multiple role-based agents.

5. Technologies Used

- Python
- Pandas

- LangGraph
- FastAPI
- CSV file

6. Implementation

A university information CSV file was created containing student names, courses, teachers, assignment deadlines, and classroom information. A search function was created to find a student's information. An answer function was then created to answer questions about the student's course, teacher, deadline, and room. LangGraph was used to connect the workflow steps.

7. Human Approval

Human approval was added for an important action such as sending an email to a teacher. The system checks whether the human approves or rejects the action. If the action is approved, it can continue. If it is rejected, the action stops. The project demonstrates the approval process but does not actually send an email.

8. Evaluation

Eight test cases were used to evaluate the system.

- Five test cases checked normal student questions.
- One test case checked an unknown student.
- One test case checked an unsupported question.
- One test case checked human approval for an important action.

Each test case achieved its expected behavior and received a 5/5 overall score.

9. Error Handling

The system handles different failure situations without crashing. If an unknown student is entered, it returns "Student not found." If the system does not understand the question, it returns a clear message instead of creating a false answer. A data-source error was also demonstrated using a missing CSV file.

10. FastAPI

A simple FastAPI endpoint was created for the Student Support Agent. It accepts a student's name and question and returns structured output containing the student name, question, and answer.

Example:

Student: Ali
Question: What is my course?
Answer: Ali is enrolled in AI course.

11. Monitoring

Basic monitoring was added to track:

- Student input
- Function calls
- Response
- Latency
- Errors
- Token usage

Token usage is currently not applicable because the current version does not use an LLM.

12. Production Monitoring

The system should monitor error rate, latency, output quality, and cost if an LLM is added in the future. The system should be reviewed if the error rate goes above 5%, latency is consistently above 1 second, or answer accuracy falls below 90%. If an LLM is added, a cost increase of more than 20% should also be reviewed.

The system should be re-evaluated once every month and whenever the CSV data, answer function, or API code is changed.

13. Limitations

The current system uses a small CSV dataset and supports only a limited number of question types. It does not currently use an LLM, so token usage and LLM cost are not applicable. The email action is only demonstrated through a human approval step and is not actually sent.

14. Future Improvements

In the future, the system can be improved by adding a larger university database, supporting more types of questions, adding authentication, connecting a real email service, and adding an LLM for more flexible question understanding. The FastAPI application could also be deployed as a real web service.

15. Conclusion

The Student Support Agent demonstrates a complete beginner-level agent system from design and implementation to evaluation, human approval, API exposure, and monitoring. The project focuses on

simple and controlled behavior, proper error handling, and safe human approval for important actions.